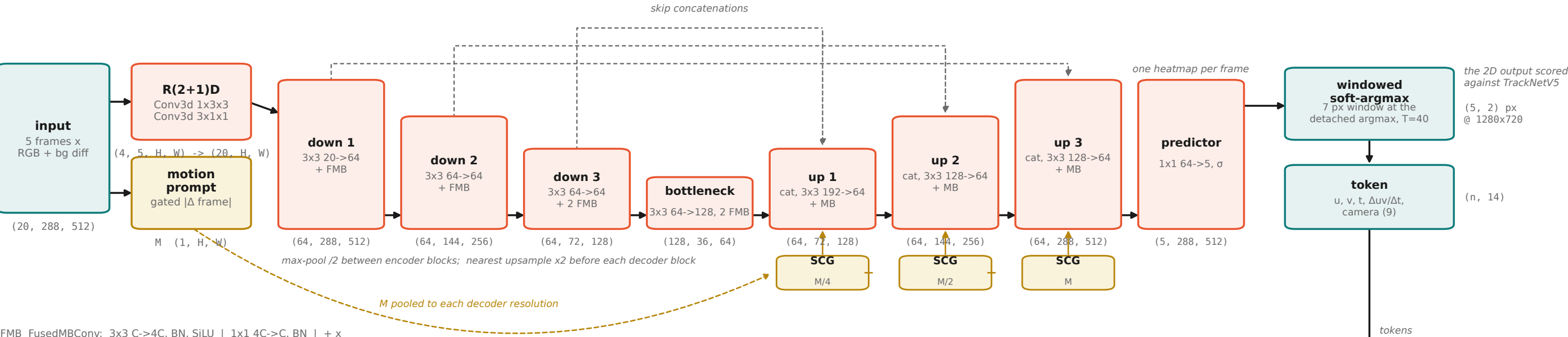


TrackNet3D

five-frame TrackNetV5-Lite -> differentiable soft-argmax -> stroke-level transformer -> closed-form geometry. Shapes: (C, H, W) at 288x512; (n, d) tokens.

A detector (TrackNetV5-Lite, 2.54 M, released weights, fine-tuned)

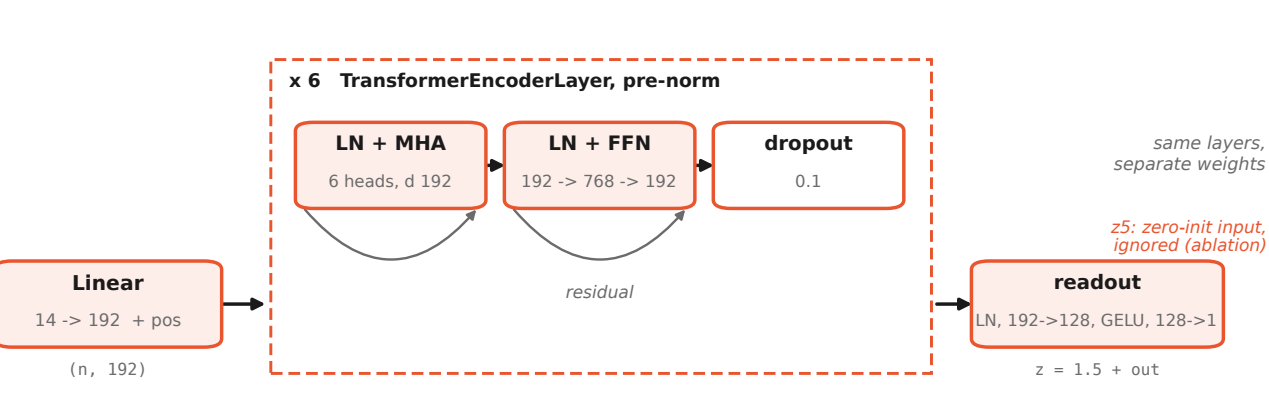


FMB FusedMBConv: 3x3 C->4C, BN, SiLU | 1x1 4C->C, BN | + x

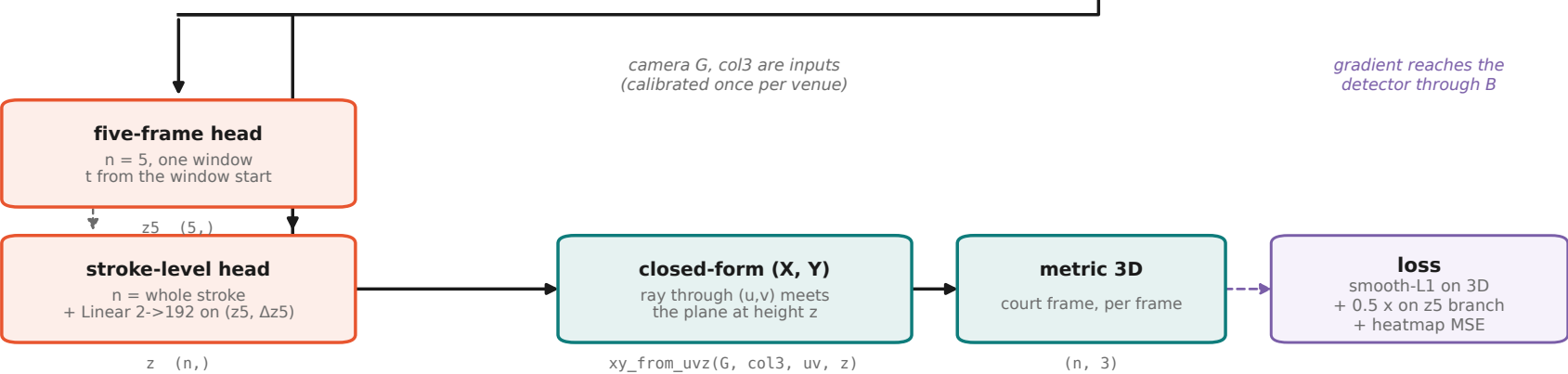
MB MBConv: 1x1 C->4C, BN, SiLU | dw 3x3, BN, SiLU | 1x1 4C->C, BN | + x

SCG spatial-channel gate: $V \cdot \sigma(\text{MLP}[\text{GAP } V, \text{GAP } M]) \cdot \sigma(3 \times 3 \text{ } M) \rightarrow \text{dw } 3 \times 3, \text{pw } 1 \times 1 \mid + V$

C lifting heads (2 x 2.80 M)



D geometry (0 parameters)



learned algebra motion gate training only

total 8.13 M parameters